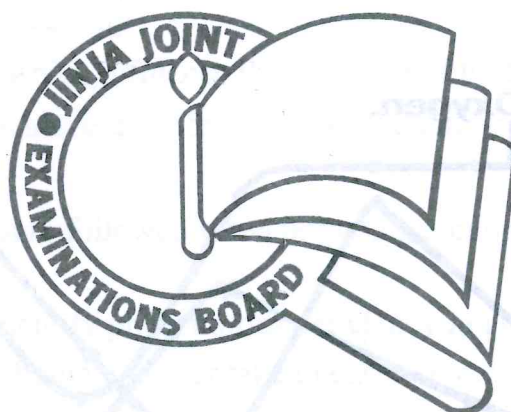


P515/2
PRINCIPLES AND
PRACTICES OF
AGRICULTURE
Paper 2
AUGUST, 2025
3 hours



JINJA JOINT EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

MOCK EXAMINATIONS – AUGUST, 2025

PRINCIPLES AND PRACTICES OF AGRICULTURE

THEORY

Paper 2

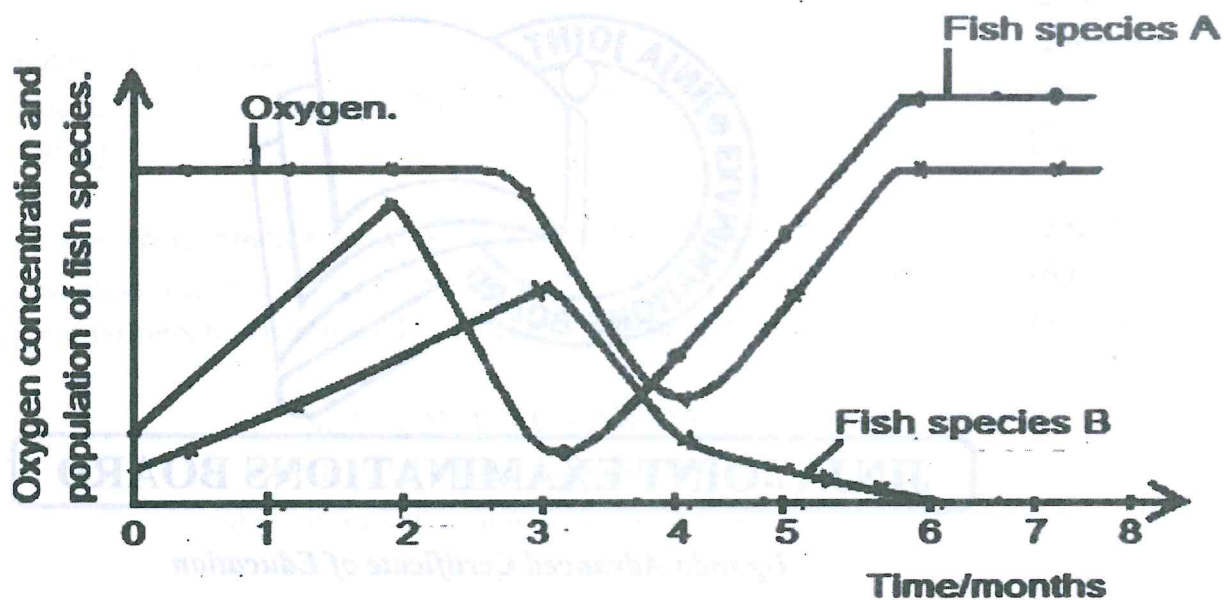
3 hours

INSTRUCTIONS TO CANDIDATES:

Answer question 1 in section A and four other questions selecting at least one from each of the sections B, C, D and E.

SECTION A: (20 MARKS)**Question 1 is compulsory**

1. Two species of fish A and B were introduced into an artificially constructed pond. Species B predate on species A. After a period of 2.7 months, a nitrogenous waste was discharged into the pond. The population of each species and the dissolved oxygen content were monitored each month. Figure 1 below shows results of the study.

**Fig.1**

- (a) Describe the changes in the population of fish species A during the experiment. (04 marks)
- (b) Compare population of fish species A and B over the period of months. (08 marks)
- (c) Suggest,
- What would happen if fish species B was removed from the pond after two months? (02 marks)
 - How the pattern of population change would be affected if the waste was not introduced into the pond. (02 marks)
- (d) Explain the changes in oxygen concentration with time from 0 to 2.5 months. (4 marks)

SECTION B (20 MARKS)

CROP PRODUCTION

Answer one question from this section.

2 (a) Define the following terms:

- (i) Non-systemic herbicides (01 mark)
- (ii) A systemic herbicide (01 mark)
- (iii) Herbicide selectivity (01 mark)

(b) Explain the factors which influence the selection of herbicides for farm use. (08 marks)

(c) How can a farmer achieve the effectiveness of a herbicide? (09 marks)

3. (a) Describe the procedure followed when preparing compost manure using indore method. (10 marks)

(b) How would you identify phosphorous deficiency in growing crops? (4 marks)

(c) What precautions should a farmer take to ensure that the farm yard manure made is of good quality? (6 marks)

SECTION C (20 MARKS)

ANIMAL PRODUCTION

Answer one question from this section.

4. (a) Describe the chemical method of dehorning a calf (12 marks)

(b) Give the reasons for carrying out the following livestock management practices.

- (i) Grooming (4 marks)
- (ii) Steaming up (4 marks)

5. (a) What is meant **brooding** in poultry production? (2 marks)

(b) Outline the causes of chick mortality during brooding. (6 marks)

(c) How can a farmer process and prepare eggs for marketing after they have been collected? (12 marks)

SECTION D (20 MARKS)

AGRICULTURAL ENGINEERING

Answer one question from this section

6. (a) Outline the specifications and requirements for:
- (i) Fish pond (4 marks)
 - (ii) Nursery bed structure (4 marks)
- (b) Describe the procedure followed when mixing good quality concrete for the foundation of a building. (12 marks)
7. (a) How can a farmer prepare a draft animal for a day's work? (5 marks)
- (b) Suggest ways a farmer can ensure good performance of an ox-plough. (6 marks)
- (c) How can farmers be encouraged to mechanize farm operations? (9 marks)

SECTION E (20 MARKS)

AGRICULTURAL ECONOMICS

Answer one question from this section

8. (a) Define the following
- (i) Opening valuation (2 marks)
 - (ii) Closing valuation (2 marks)
- (b) State the features of a profit and loss account (5 marks)
- (c) (i) What factors determine the type of record kept on the farm? (5 marks)
- (ii) Give the importance of keeping inventory records (6 marks)
9. (a) What is meant by?
- (i) Land adjudication (2 marks)
 - (ii) Land registration (2 marks)
- (b) (i) State the benefits of having a land title deed to a farmer. (6 marks)
- (ii) Outline the features of a land titled deed (5 marks)
- (c) Give the causes of land fragmentation (5 marks)